无人系统设计 课程作业04

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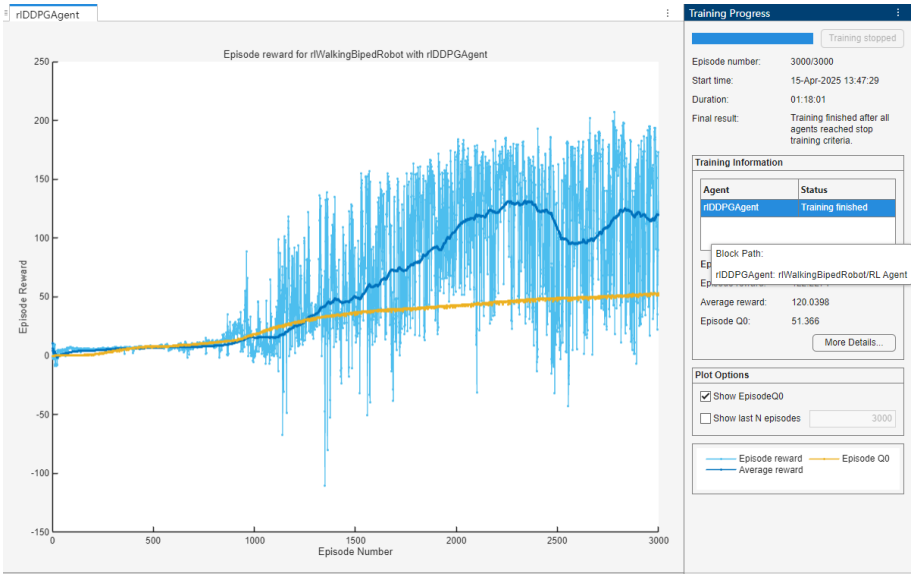
1 作业1

1.1 硬件环境

CPU为Intel i5-12400F，GPU为RTX 2060s，内存32G。

1.2 训练过程

下图为训练结果。



	rlDDPGAgent
Status	Training finished
Episode number	3000
Episode reward	122.2274
Episode steps	360
Total agent steps	672126
Average reward	120.0398
Average steps	310.76
Episode Q0	51.366
Averaging window length	250
Training stopped by	EpisodeCount
Training stopped at	3000

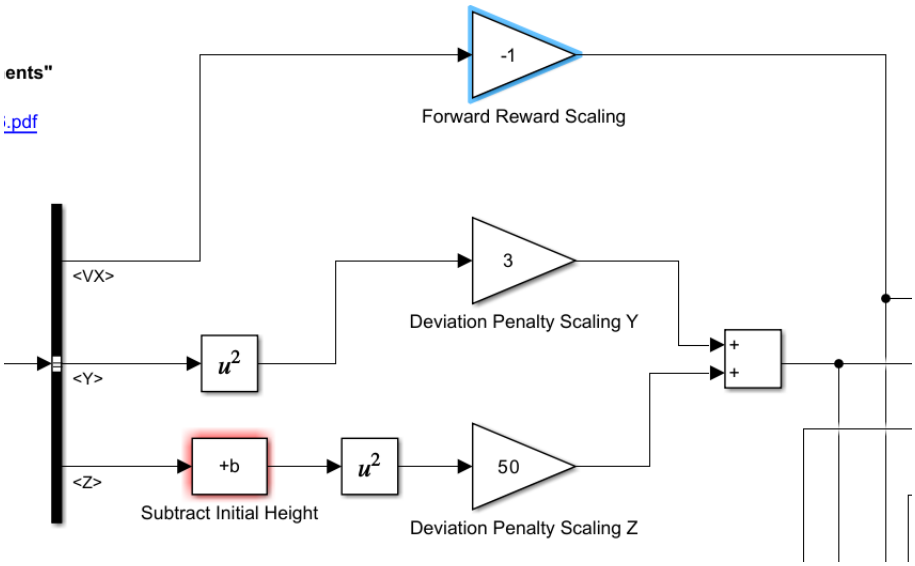
1.3 实验心得

奖励函数与机器人在正方向的位移正相关，与y、z方向位移，关节扭矩等因素负相关，当机器人摔倒或横向偏移过大时，会终止训练。由此可以训练机器人向正方向移动。DDPG的最终效果可能不如TD3，但是训练速度要优于后者，为了节省时间，我选择前者作为训练算法。最终训练出来的结果如上图，机器人行走视频见文件夹内。

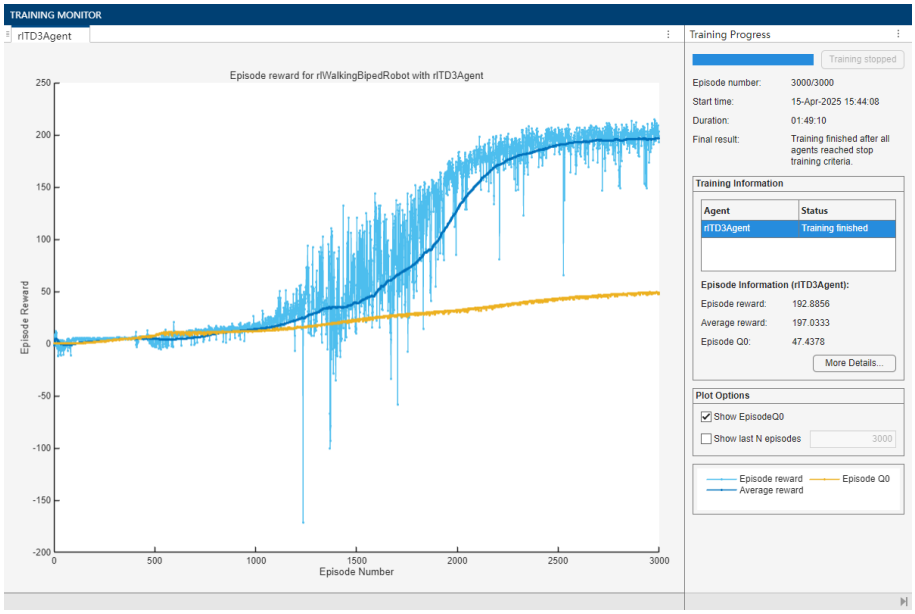
2 作业2

2.1 训练过程

训练机器人倒退行走，只需要将正方向视为反方向即可，因此只需修改奖励函数中关于x方向的部分，如下图，修改V_x的参数为-1。



下图为训练结果。



	rlTD3Agent
Status	Training finished
Episode number	3000
Episode reward	192.8856
Episode steps	400
Total agent steps	717852
Average reward	197.0333
Average steps	399.804
Episode Q0	47.4378
Averaging window length	250
Training stopped by	EpisodeCount
Training stopped at	3000

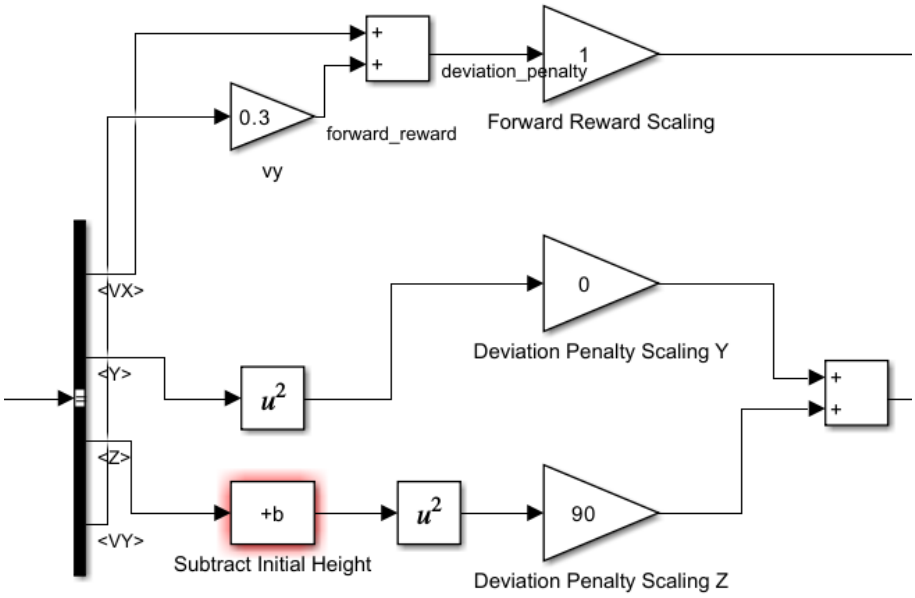
2.2 实验心得

只需要适当修改奖励函数，就可以引导机器人向其他方向行走。为了提高训练结果的质量和稳定性，这次训练选择使用TD3算法进行训练。

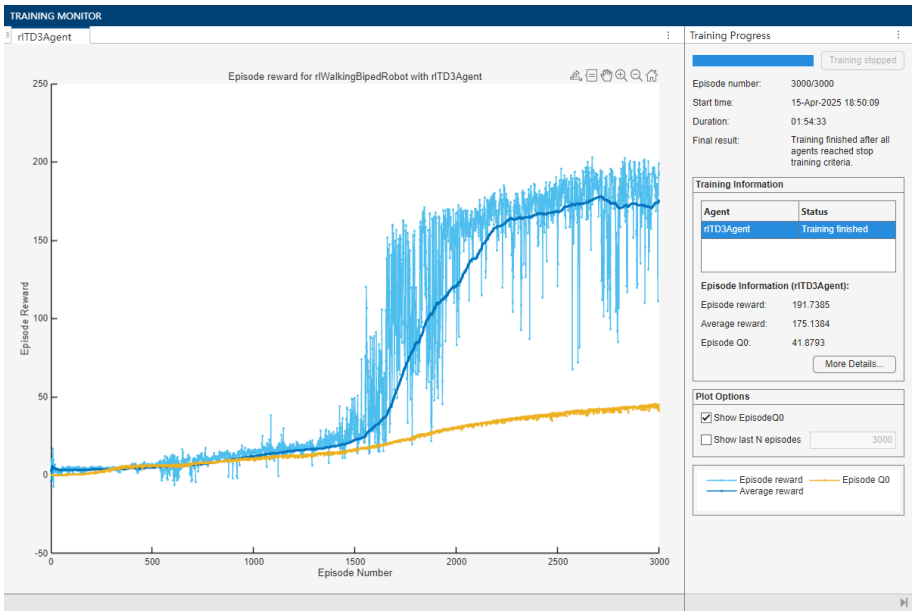
3 作业3

3.1 训练过程

设置训练目标为让机器人斜着走。修改奖励函数，首先把y方向的惩罚去除，其次增加一部分Vy的奖励以鼓励机器人向左前方走。由于机器人本身结构限制，不能让y方向的倾斜角度过大，否则会导致机器人倒下，最终设定Vy奖励系数为0.3。最后，由于机器人斜着走更需要稳定性，因此加大z方向的惩罚，确保机器人可以稳步行走。


+
+
deviation_penalty
1
Forward Reward Scaling
0.3
forward_reward
vy
<VX>
<Y>
<Z>
<VY>
0
Deviation Penalty Scaling Y
u^2
+b
Subtract Initial Height
u^2
90
Deviation Penalty Scaling Z

下图为训练结果。


TRAINING MONITOR
rlTD3Agent
Episode reward for rlWalkingBipedRobot with rlTD3Agent
Episode Reward
Episode Number
Training Progress
Training stopped
Episode number: 3000/3000
Start time: 15-Apr-2025 18:50:09
Duration: 01:54:33
Final result: Training finished after all agents reached stop training criteria.
Training Information
Agent
Status
rlTD3Agent
Training finished
Episode Information (rlTD3Agent):
Episode reward: 191.7385
Average reward: 175.1384
Episode Q0: 41.8793
More Details...
Plot Options
Show EpisodeQ0
Show last N episodes
3000
Episode reward
Episode Q0
Average reward

	rlTD3Agent
Status	Training finished
Episode number	3000
Episode reward	191.7385
Episode steps	400
Total agent steps	590464
Average reward	175.1384
Average steps	382.956
Episode Q0	41.8793
Averaging window length	250
Training stopped by	EpisodeCount
Training stopped at	3000

3.2 实验心得

适当修改奖励函数，即可正确引导机器人斜着走。视频见附件文件夹内。